

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

The electrode potential of M^{2+}/M of 3d-series elements shows positive value for:

- (1) Zn
- (2) Co
- (3) Fe
- (4) Cu

Q2: 24 Feb (Shift 2) - Numerical

The magnitude of the change in oxidising power of the MnO_4^-/Mn^{2+} couple is $x \times 10^{-4}$ V, if the H^+ concentration is decreased from 1M to 10^{-4} M at $25^\circ C$. (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H^+ concentration). The value of x is _____ (Rounded off to the nearest integer) [Given

$$\frac{2303RT}{F} = 0.059]$$

Q3: 25 Feb (Shift 2) - Numerical

Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution.

(Assuming fixed $[Cu^{2+}]$ and $P_{NO} = P_{NO_2}$), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is 10^x M. The value of 2x is _____ (Rounded-off to the nearest integer)

[Given : $E^0_{Cu^{2+}/Cu} = 0.34$ V, $E^0_{NO_3^-/NO} = 0.96$ V, $E^0_{NO_3^-/NO_2} = 0.79$ V and at 298 K, $\frac{RT}{F}(2.303) = 0.059]$

Q4: 26 Feb (Shift 1) - Numerical

Consider the following reaction



The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is _____

Q5: 26 Feb (Shift 2) - Numerical

Questions with Answer Keys

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Emf of the following cell at 298 K in V is $x \times 10^{-2}$. $\text{Zn} | \text{Zn}^{2+} (0.1\text{M}) || \text{Ag}^+ (0.01\text{M}) | \text{Ag}$

The value of x is _____ (Rounded off to the nearest integer) [Given

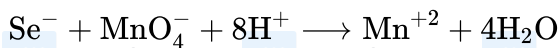
$$: E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76 \text{ V}; E_{\text{Ag}^+/\text{Ag}}^0 = +0.80 \text{ V}; \frac{2.303RT}{F} = 0.059]$$

Answer Key**Q1 (4)****Q2 (3776)****Q3 (1)****Q4 (25)****Q5 (147)**

Q1 (4)

(A) Zn -0.76 (B) CO -0.28 (C) Fe -0.44 (D) Cu $+0.34$

Q2 (3776)



$$Q = \frac{[\text{Mn}^{+2}]}{[\text{H}^+]^8 [\text{MnO}_4^-]} \Rightarrow E_1 = E^\circ - \frac{0.059}{5} \log(Q_1)$$

$$E_2 = E^\circ - \frac{0.059}{5} \log(Q_2) \Rightarrow E_2 - E_1 = \frac{0.059}{5} \log\left(\frac{Q_1}{Q_2}\right)$$

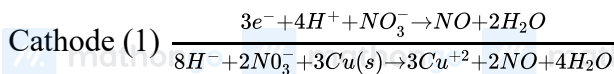
$$= \frac{0.059}{5} \log \left\{ \frac{[\text{H}^+]_{\text{II}}}{[\text{H}^+]_{\text{I}}} \right\}^8 \Rightarrow = \frac{0.059}{5} \log \left(\frac{10^{-4}}{1} \right)^8$$

$$(E_2 - E_1) = \frac{0.059}{5} \times (-32) \Rightarrow |(E_2 - E_1)| = 32 \times \frac{0.059}{5} = x \times 10^{-4}$$

$$= \frac{32 \times 590}{5} \times 10^{-4} = x \times 10^{-4} \Rightarrow = 3776 \times 10^{-4} \quad x = 3776$$

Q3 (1)

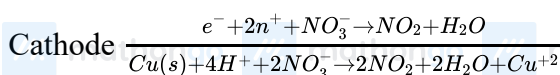
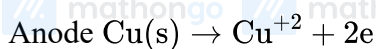
Anode



$$Q = \frac{[\text{Cu}^{+2}]^3 \times (p_{\text{NO}})^2}{[\text{NO}_3^-]^2 [\text{H}^+]^8}$$

$$E_{\text{cell}}^0 = 1.3$$

$$E_{\text{cell}} = 1.3 - \frac{0.059}{6} \log \frac{(\text{Cu}^{+2})^3 (p_{\text{NO}})^2}{(\text{NO}_3^-)^2 \times (\text{H}^+)^8} \dots (1)$$



Hints and Solutions

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$$E_{\text{cell}}^0 = 1.13$$

$$Q = \frac{(Cu^{+2})(p_{NO_2})^2}{(NO_3^-)^2(H^+)^4}$$

$$E_{\text{cell}} = 1.13 - \frac{0.059}{2} \log \frac{(Cu^{+2})(p_{NO_2})^2}{(NO_3^-)^2(H^+)^4}$$

$$E_{\text{cell}_1} = E_{\text{cell}_2}$$

$$1.3 - \frac{0.059}{6} \log(Q_1) = 1.13 - \frac{0.059}{2} \log(Q_2)$$

$$0.17 = \frac{0.059}{6} \{ \log(Q_1) - 3 \log(Q_2) \}$$

$$= \frac{0.059}{6} \left\{ \log \frac{(Cu^{+2})^3 \times (p_{NO})^2 \times (NO_3^-)^6 (H^+)^{12}}{(NO_3^-)^2 (H^+)^8 \times (Cu^{+2})^3 \times (p_{NO_2})^6} \right\}$$

$$= \frac{0.059}{6} \left\{ \log \frac{[NO_3^-]^4 [H^+]^4}{(p_{NO_2})^4} \right\}$$

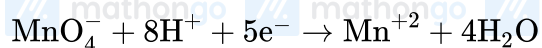
$$0.17 = \frac{0.059}{6} \times 8 \log(HNO_3)$$

$$\log(HNO_3) = 2.16$$

$$[HNO_3] = 10^{2.16} = 10^x$$

$$x = 2.16 \Rightarrow 2x = 4.32 \approx 4$$

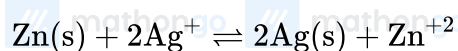
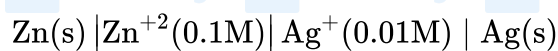
Q4 (25)



1 mole of MnO_4^- require 5 faraday charge

5 moles of MnO_4^- will require 25 faraday charge.

Q5 (147)



$$E^0 = 0.80 + 0.76 = 1.56; Q = \left\{ \frac{Zn^{2+}}{(Ag^+)^2} \right\}$$

$$E = E^0 - \frac{0.059}{n} \log(Q)$$

$$E = 1.56 - \frac{0.059}{2} \log \left[\frac{0.1}{(0.01)^2} \right]$$

$$E = 1.56 - \frac{0.059}{2} \log[(10)^3]$$

$$E = 1.4715 = 147.15 \times 10^{-2} \text{ volt}$$

$$= x \times 10^{-2}$$

$$X = 147.15 \approx 147 \text{ Ans}$$